Optical Delay Line

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1 Introduction

This document describes an optical delay line module built from Thorlabs XR25 series components. The motivation for designing this delay line was to achieve photon indistinguishability in time for the Hong–Ou–Mandel (HOM) interferometer. By changing the optical path length, the arrival time of one photon can be adjusted until it matches the other photon.

The delay line is designed as a standalone setup, so it can also be removed or reused in other experiments that require an adjustable optical path. The main components are a 250 mm dovetail rail, a parallel mounting adapter, a 25 mm translation stage, a mirror plate, and two prism mounts. Figure 1 shows the CAD assembly of the proposed delay line.

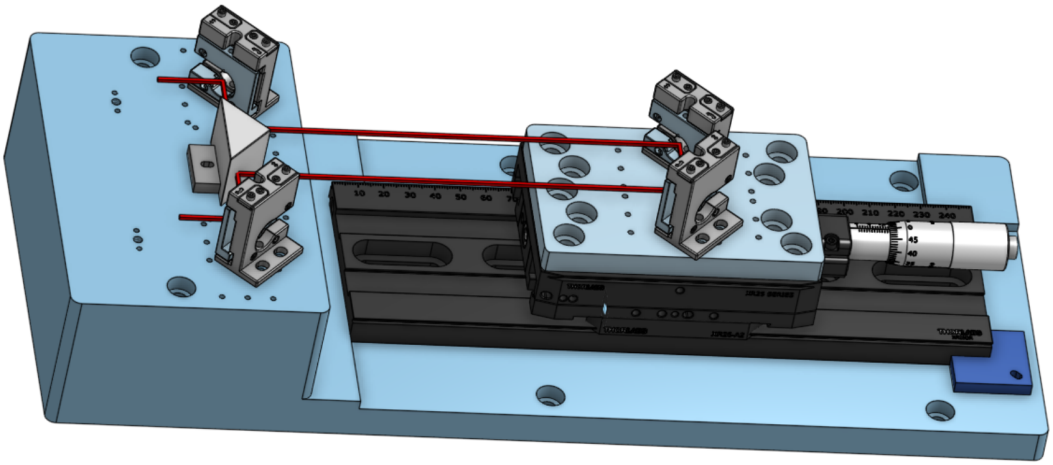


Figure 1: Proposed Delay Line

Link to CAD File

2 Base Sketch

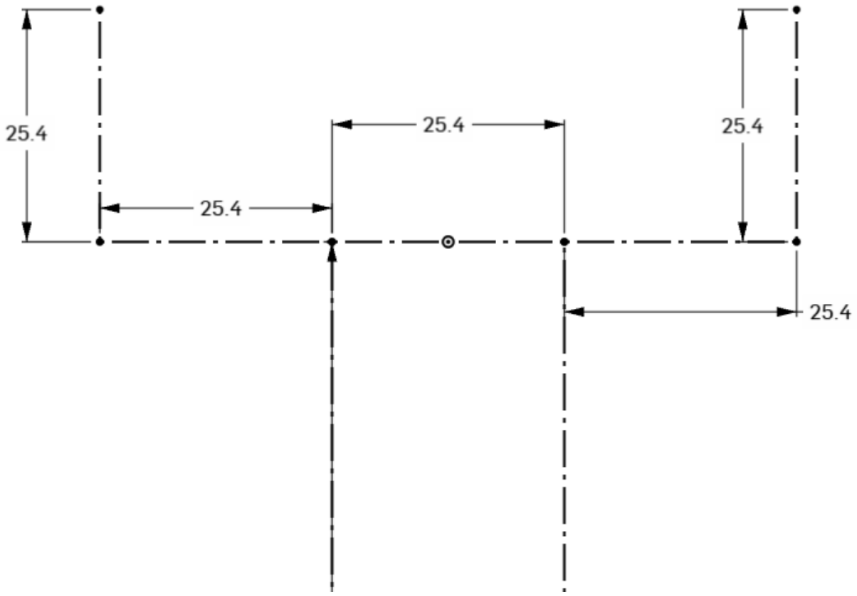


Figure 2: Base Sketch

Figure 2 shows the basic optical path used to place the components of the delay line. Each solid dot corresponds to an optic on the mount, while the circled dot ($\odot$) at the center marks the sketch origin. The dash-dot lines represent the optical axes followed by the beam through the delay line.

The 25.4 mm reference, measured from the midpoint origin, sets how the prism is positioned so that the beam going toward the translation stage and the beam returning from it remain one inch apart. The remaining components are then placed relative to these optical axes. Keeping these references in the base sketch makes it easier to position the prism mounts, rail, and mirror plate from the same geometry.

2.1 Baseplate Heights

The baseplate uses two different extrusion heights so that the optics on both sides of the delay line remain at the same final height. The raised section on the left, which holds the mirrors, collimators, and prisms, is 2.5 inches high. Since the optical components add approximately another half inch, the optical axis is placed at a total height of about 3 inches.

The right side holds the dovetail rail, parallel mounting adapter, translation stage, and the 10 mm mirror plate. The rail, adapter, and translation stage together have a height of approximately 40 mm. Therefore, the extrusion underneath this assembly is made lower so that the mirror plate reaches the same height as the left side. Its height is

$$2.5\text{ in} - 40\text{ mm} - 10\text{ mm} = 63.5\text{ mm} - 40\text{ mm} - 10\text{ mm} = 13.5\text{ mm}. \tag{1}$$

This makes the right-side extrusion approximately 13.5 mm high. After the rail, adapter, translation stage, and mirror plate are added, the top of the mirror plate is at the same 2.5-inch reference height as the raised section on the left. The mounted optics then bring the beam to the intended 3-inch optical height.

3 Mirror Mounts

Two mirror-mount families are currently supported: the rear-loading Opto-Sigma VMHV-12.7L-R and the front-loading Thorlabs Polaris-K05VS2. In both cases, each mount's hole pattern is tied back to the fixed point of incidence defined in the base sketch, and the only difference between mounts is the offset used to place the mount's reference edge relative to that point. For the rear-loading Opto-Sigma mount, the mirror always sits at a fixed distance behind the front face, so the offset is independent of the optic: the pin-hole row is referenced 3.5 mm and the screw-hole row 15.5 mm below the point of incidence. For the front-loading Polaris mount, the mirror thickness sets where the surface lands, so the reference edge is offset accordingly; it is drawn for two thicknesses, giving a $+4.4$ mm offset for a 3 mm mirror and a -4.6 mm offset for a 12 mm mirror, the sign flip reflecting that the thicker mirror pushes the reference edge to the opposite side of the point of incidence. At present the design only accounts for these two mounts — the Opto-Sigma mount and the Thorlabs Polaris mount for the two mirror thicknesses. For a fuller explanation of how these mirror mounts are set up and referenced, see the HOM Documentation.

4 Prism Mount

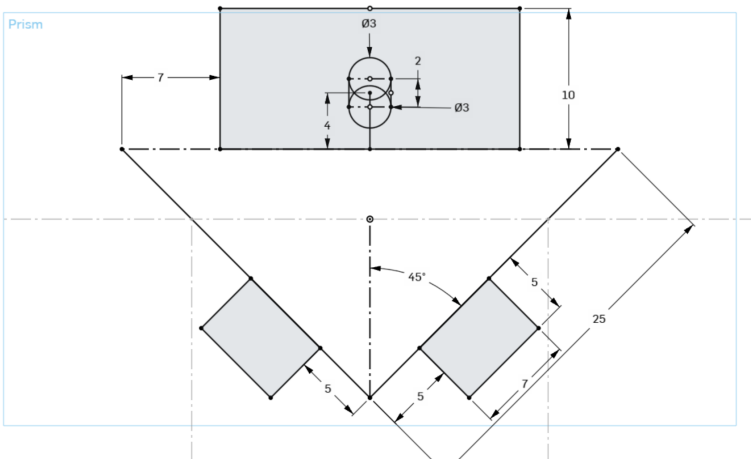


Figure 3: Prism Mount Sketch

Figure 3 shows the sketch used for mounting the prism on the base module. The prism is positioned between the two rectangular supports shown along the angled sides. These supports are extrusions made directly on the base and provide surfaces for the prism to rest against. Their placement follows the 45° prism geometry and keeps the prism in the required position relative to the incoming and outgoing beam paths.

The rectangular part above the prism is the locking piece. Unlike the side supports, this is a separate part that is screwed into the base after the prism is placed in position. Tightening this piece holds the prism against the two side extrusions and keeps it from moving. The lock can be unscrewed when the prism needs to be removed or replaced.

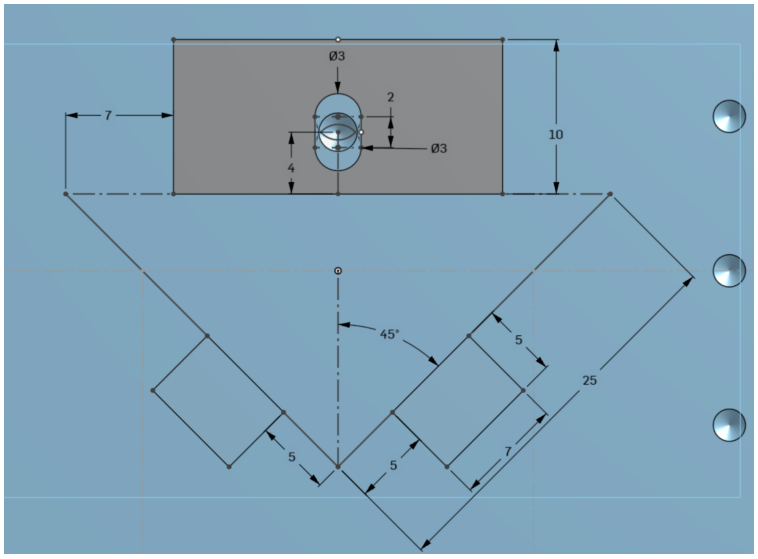


Figure 4: CAD Rendered Prism Mount View

Figure 4 shows the same prism mount in the CAD rendered base. The two side extrusions are part of the base module, while the upper rectangular piece is the removable lock. Together, these features hold the prism in its intended position while keeping the mounting method simple.

5 XR25DR: 3” Dovetail Rail, 250 mm Long

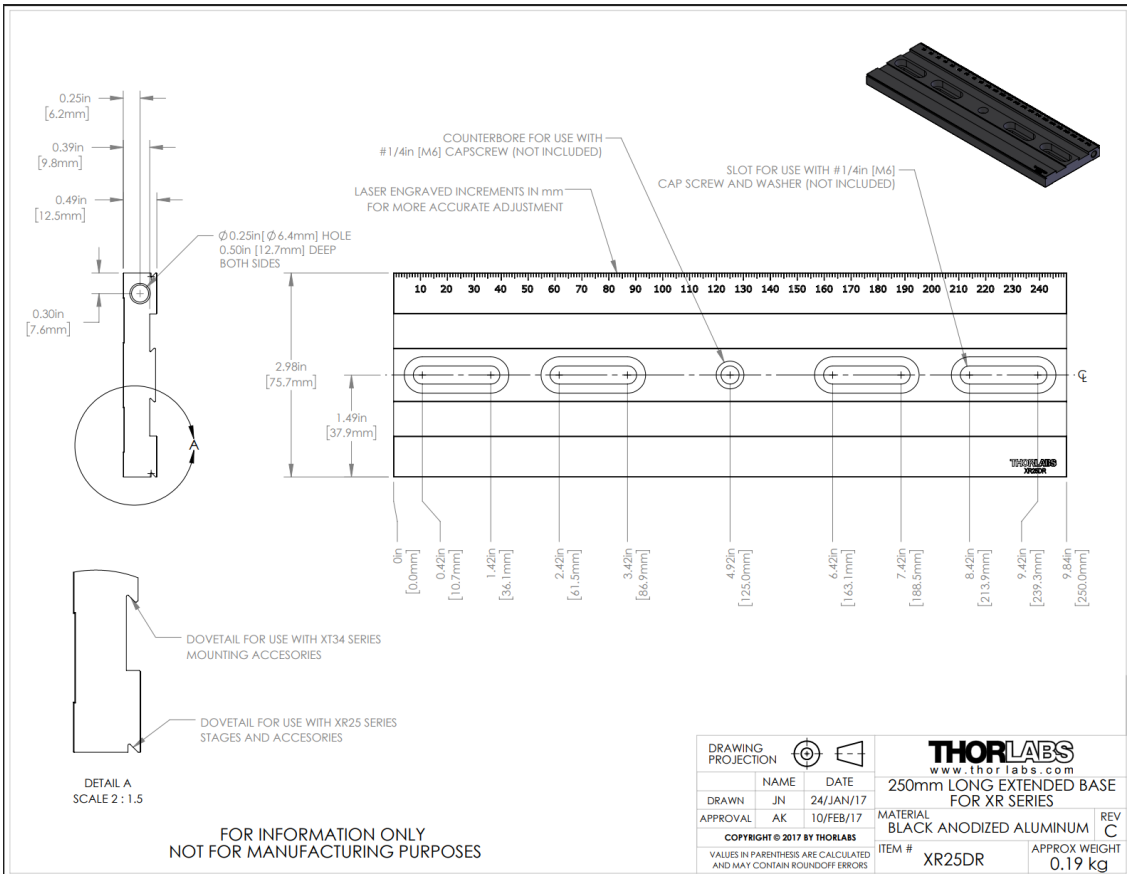


Figure 5: XR25DR Official CAD Drawings

Link to the product

The XR25DR is the main rail used for the delay line and provides the base for the moving assembly. It has a 3-inch dovetail interface along its 250 mm length, which allows the parallel mounting adapter and translation stage to be moved along the rail before being locked into position. This gives a coarse adjustment of where the moving part of the delay line sits.

Figure 5 shows the official Thorlabs CAD drawing used as the dimensional reference. The rail also contains mounting holes along its length for securing it to the custom base. From the drawing, the spacing between consecutive mounting-hole locations is a center-to-center distance of 2 inches.

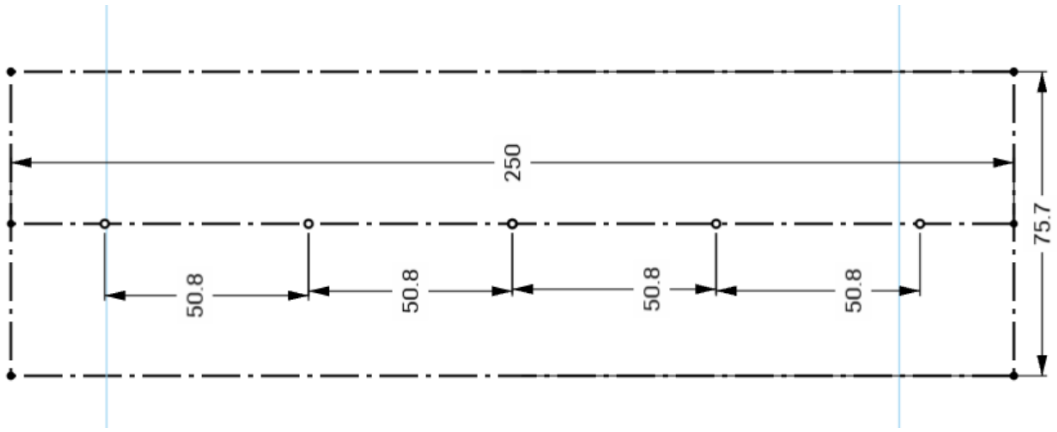


Figure 6: Sketch for Dovetail Rail Mount

Figure 6 shows the sketch used to place the dovetail rail on the custom mount. Each circular dot corresponds to a mounting hole that fixes the rail to the base. The position of these holes keeps the rail aligned with the optical path defined in the base sketch.

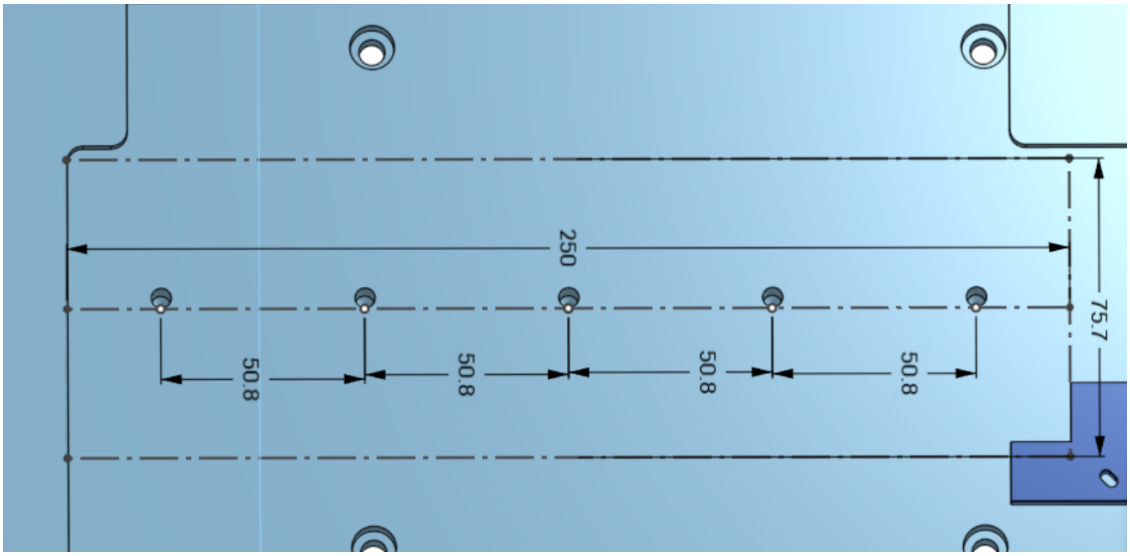


Figure 7: Rendered View for Dovetail Rail Mount

Figure 7 shows the resulting CAD geometry. Additional guide features and extrusions were added along the mount to help keep the rail straight during installation and reduce placement error. A lock at the bottom-right side of the mount is used to hold the rail firmly against these supports after it is placed in position.

6 XR25-A2: Parallel Mounting Adapter with Male 3” Dovetail and Female 3” Dovetail

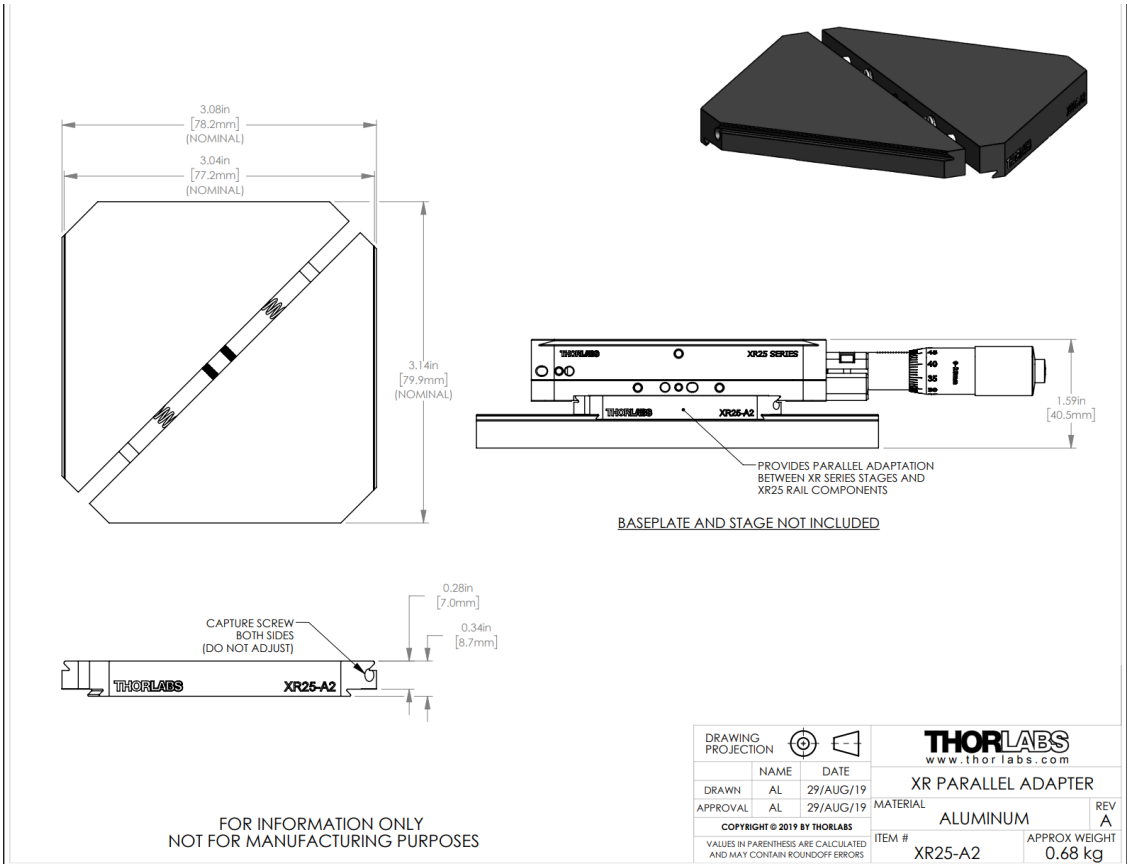


Figure 8: XR25-A2 Official CAD Drawings

Link to the product

The XR25-A2 is the adapter used between the dovetail rail and the translation stage. One side connects to the XR25DR rail, while the opposite side provides the interface for mounting the translation stage. This allows the translation stage to sit parallel to the rail while still being movable along the rail for coarse positioning.

Once the desired position is reached, the adapter can be locked onto the rail. The translation stage then provides the smaller controlled motion used during operation of the delay line. In this way, the rail and adapter provide the overall placement of the moving assembly, while the translation stage provides the actual delay adjustment.

7 XR25C: 25 mm Translation Stage

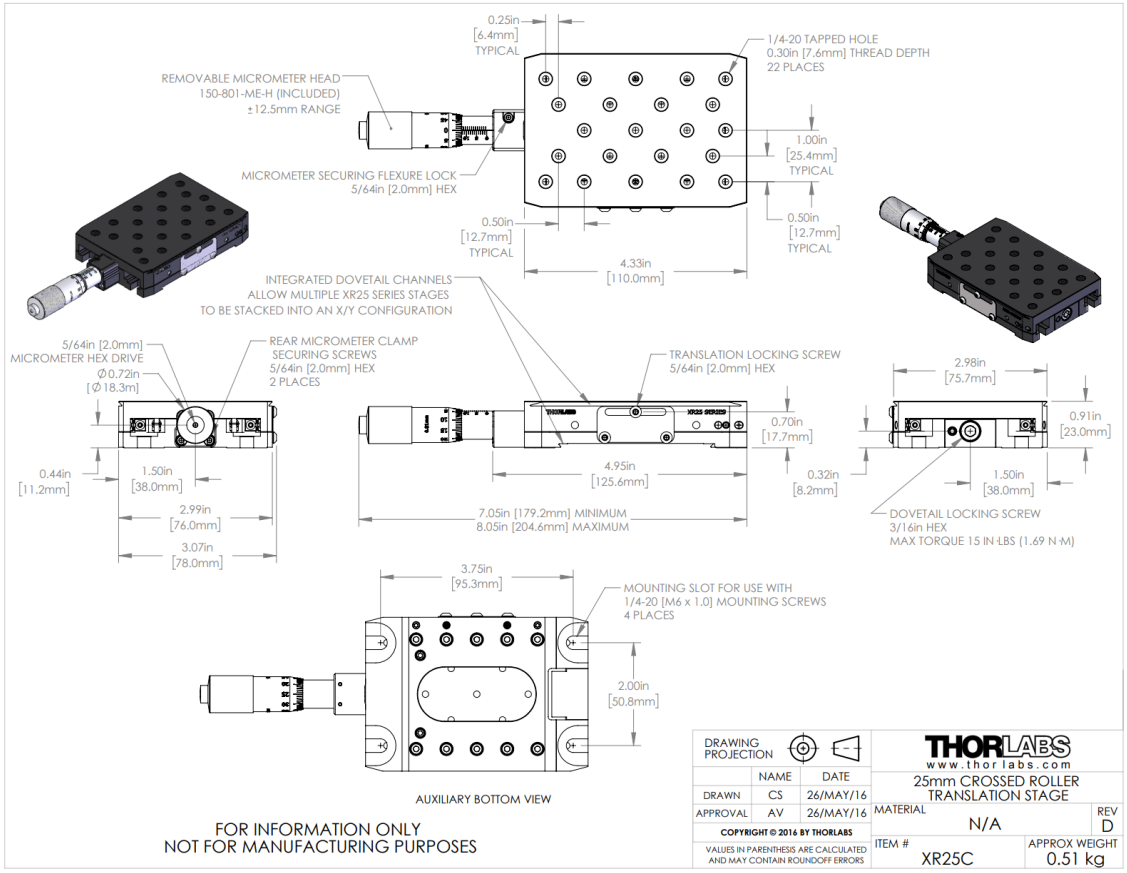


Figure 9: XR25C Official CAD Drawings

Link to the product

The XR25C translation stage provides the controlled linear motion of the delay line. It has 25 mm of travel, and the mirror plate is mounted directly on top of it. Moving the stage therefore moves both mirrors together and changes the distance traveled by the beam before it returns to the prism.

The micrometer at the end of the stage is used for fine adjustment of this position. The top surface contains the threaded mounting holes used to attach the custom mirror plate. Since the complete mirror assembly moves with the stage, the spacing between the two mirrors remains fixed while the optical path length is adjusted.

8 Assembly

The delay line is assembled in several steps so that the rail, translation stage, and optical components can be positioned separately before the complete system is put together. The parallel mounting adapter is first attached to the dovetail rail, followed by the translation stage. The mirror plate is then mounted on top of the stage. The subsections below show these steps individually before showing the completed assembly.

8.1 Dovetail with Parallel Mounting Adapter



Figure 10: Assembly for Parallel Adapter with Dovetail Rail

Figure 10 shows the XR25-A2 adapter installed on the dovetail rail. The adapter slides along the rail and can be moved to the approximate location required for the delay line before being locked in place. This provides the coarse positioning of the moving portion of the setup and leaves the translation stage to make the smaller adjustments during operation.

8.2 Translation Stage with Parallel Mounting Adapter



Figure 11: Assembly for Parallel Adapter with Translation Stage

Figure 11 shows the translation stage mounted on top of the parallel adapter. The stage is oriented with the micrometer facing outward so that it remains accessible after the delay line is assembled. The adapter keeps the stage aligned with the dovetail rail, while the stage provides the motion that will later move the mirror plate.

8.3 Dovetail + Parallel Mounting + Translation Stage

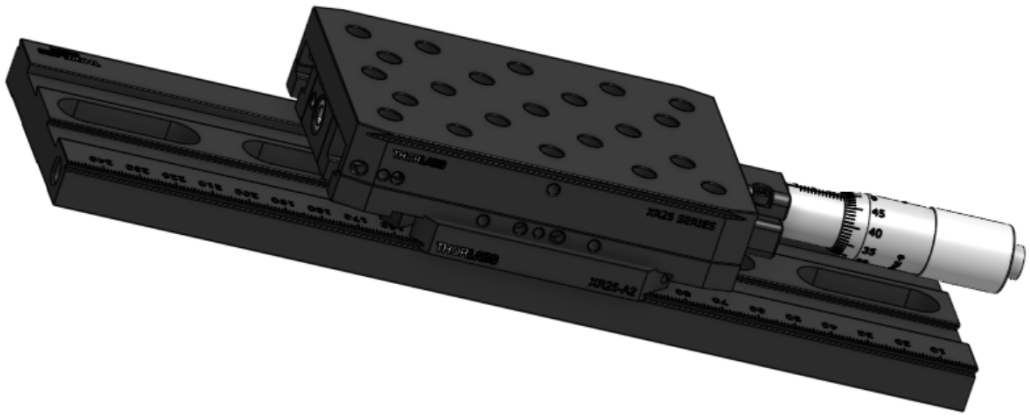


Figure 12: Assembly for Parallel Mounting Plate, Dovetail, and Translation Stage

Figure 12 shows the rail, parallel adapter, and translation stage assembled together. At this point, the main moving portion of the delay line is complete. The adapter can be positioned along the rail for coarse placement, while the micrometer on the translation stage provides the smaller controlled motion. The mirror plate is mounted on top of this assembly in the next step.

8.4 Mirror Plate

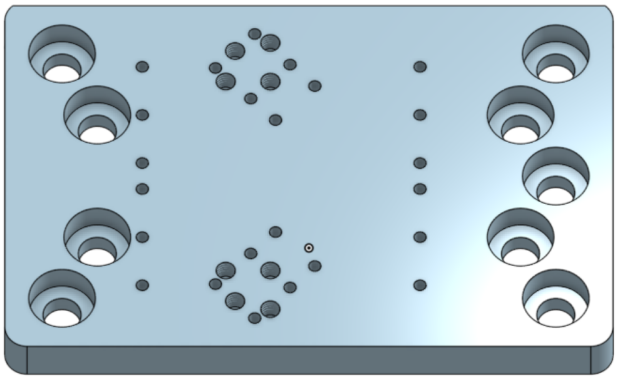


Figure 13: Mirror Plate

Figure 13 shows the custom mirror plate that sits on top of the translation stage. The plate is 10 mm thick and provides the mounting locations for the two mirror mounts used to fold the beam back toward the prism. The two mirror

positions are separated by one inch so that the outgoing and returning beam paths remain parallel and one inch apart.

The larger counterbored holes near the corners are used to fasten the plate to the translation stage. The smaller holes positioned before and after the mirror locations can be used for alignment irises. These provide convenient reference points when aligning the beam through the delay line.

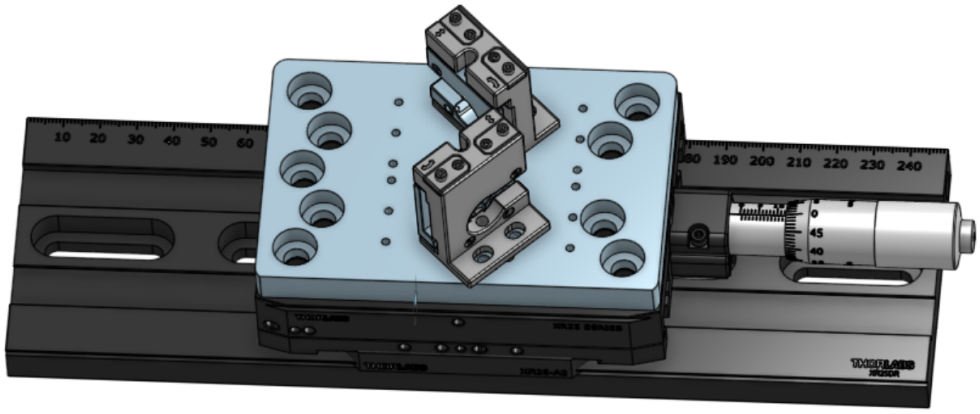


Figure 14: Mirror Plate Assembly

Figure 14 shows the mirror mounts installed on the completed plate. The two mirrors form the return path of the delay line, sending the beam back toward the second prism. Since both mirrors are attached to the same plate, they move together when the translation stage is adjusted. This changes the optical path length while keeping the spacing between the outgoing and returning beams unchanged.

8.5 Final Assembly

The final assembly combines the two prism mounts, dovetail rail, parallel adapter, translation stage, and mirror plate into the complete optical delay line. The first prism redirects the incoming beam toward the two mirrors mounted on the translation stage. After reflecting from the mirror pair, the beam travels back parallel to the incoming path and reaches the second prism.

Because the mirror plate is attached to the translation stage, turning the micrometer moves both mirrors together. Moving the mirrors farther from the prisms increases the distance traveled by the beam, while moving them closer decreases it. This provides the adjustable optical path needed to match the photon arrival times in the HOM interferometer.

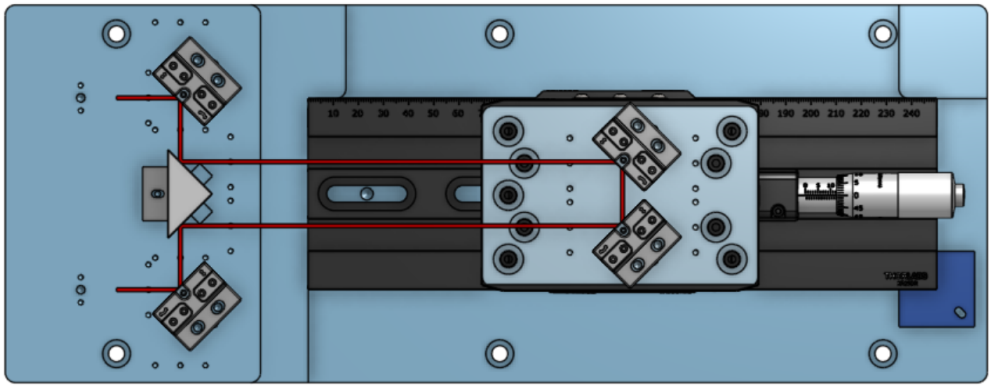


Figure 15: Assembly Top View

Figure 15 shows the completed delay line from above. The outgoing and returning beam paths can be seen running parallel between the prism mounts and the mirror plate. These two paths remain one inch apart throughout the delay line. The top view also shows how the rail and translation stage are placed relative to the optical path while keeping the micrometer accessible for adjustment.

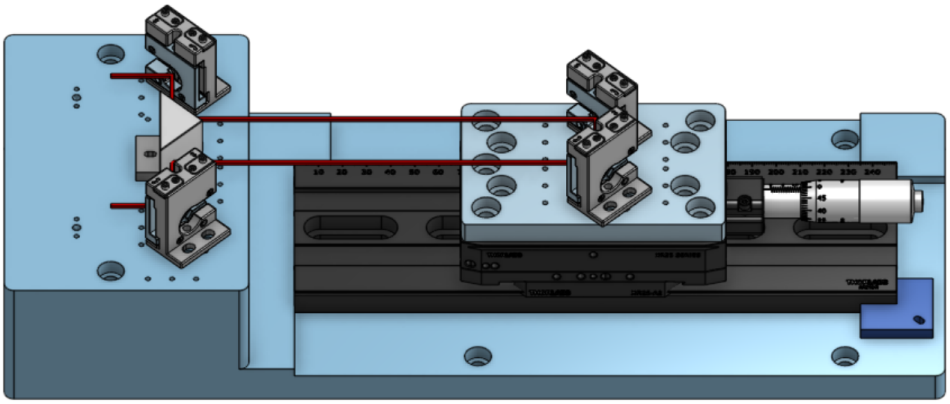


Figure 16: Assembly Side View

Figure 16 shows the completed assembly from the side. This view is mainly useful for showing the relative heights of the prism, mirror plate, and translation-stage assembly. The components are positioned so that the beam remains at the same height as it travels from the prism to the mirrors and back through the delay line.

Link to CAD File